

Chris Dollo

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Education

University of Maryland, Baltimore County

Bachelor of Science in Computer Science

Jan 2022 – Dec 2025

- Relevant Coursework: Data Structures, Algorithms, Software Engineering, Machine Learning
- Awards: President's list (x4), Dean's list (x3), Cyber Scholar

Research Experience

Undergraduate Research Assistant

Jan 2024 – Present

UMBC Vinjamuri Lab

Catonsville, MD

- Co-authored a Nature Scientific Reports paper introducing a novel framework for understanding hand gestures using structured movement primitives derived from Indian classical dance (Bharatanatyam).
- Designed and implemented a real-time Python pipeline for gesture analysis and reconstruction using OpenCV and Google MediaPipe.
- Extracted and modeled joint landmarks and Gaussian-based joint angular velocities from over 50 mudras and natural hand grasps to derive “syllables of motion.”
- Achieved 95.78% reconstruction accuracy for natural grasps and 92.99% for mudras, outperforming baseline models (88.92% and 82.51%, respectively).
- Developed deep learning models for classifying hand gestures from 24-channel electromyographic (EMG) sensor data, including preprocessing, feature extraction, and model training.
- Used interpolation and feature engineering to preprocess an EMG dataset, which improved model performance and stability.
- Mentored 3 high school students on research projects, guiding them through data analysis and experimental design.

Publications

- Olikkal, P., Dollo, C., Ajendla, A. et al. Reconstructing hand gestures with synergies extracted from dance movements. Sci Rep 15, 41670 (2025). <https://doi.org/10.1038/s41598-025-25563-7>

Work Experience

Machine Learning Intern

May 2025 – Present

Verizon

Catonsville, MD

- Designed and deployed an end-to-end ML pipeline using YOLOv8 to detect birds and nests on cell towers, achieving 94% mAP on a dataset of 14,000+ images.
- Preprocessed diverse datasets from Roboflow and Hugging Face, improving model robustness through data augmentation (rotation, scaling, brightness/contrast, saturation).
- Optimized training workflows and inference performance, reducing false detections by 20% and enhancing model reliability for real-time inspection use cases.
- Developed automated unit and integration tests to validate model outputs within the production workflow.

Personal Tutor for ROTC students

Aug 2023 - Dec 2024

UMBC Dean of Natural Mathematical Sciences

Catonsville, MD

- Lead 5+ one-on-one tutoring sessions per week with ROTC students, focusing on Algebra I to Calculus II
- Helped 7 ROTC students improve their grades by an average of 15% by developing personalized study schedules and providing studying resources

Teaching Assistant for Calculus I

Aug 2023 - May 2024

UMBC Department of Mathematics and Statistics

Catonsville, MD

- Led 2 weekly discussion sessions for 2 classes of 30 students each, improving their understanding of Calculus I concepts
- Graded and recorded grades for 500+ quizzes, ensuring timely feedback on the academic progress of each student
- Maintained 100% attendance tracking for review sessions to monitor and encourage student engagement.

Teaching Assistant for Computer Science II

Jan 2023 - May 2024

UMBC Department of Computer Science & Electrical Engineering

Catonsville, MD

- Provided hands-on debugging support during labs, helping students fix segmentation faults, pointer bugs, memory-management issues, and logic errors in C++ programs.
- Designed and delivered interactive mini-lessons using analogies, code walk-throughs, and live-coding demos to clarify difficult concepts like linked lists, dynamic memory, and runtime complexity.
- Graded 250+ programming assignments and exams, ensuring consistent evaluation and actionable feedback that improved student code quality across the semester.
- Held 4–6 office hours per week, offering individualized guidance on problem-solving strategies, debugging workflows, and preparing students for technical exams.
- Collaborated with course instructors to update assignment specifications and grading rubrics, ensuring fairness, clarity, and alignment with learning objectives.

Physics 122 SI PASS Leader

July 2023 - Aug 2023

Academic Success Center

Catonsville, MD

- Facilitated Peer Assisted Study Sessions (PASS) to support traditionally challenging courses, promoting collaborative learning and effective study strategies.
- Led group exercises that helped students explain concepts to each other, improving class-wide understanding of circuits, electromagnetism, the right-hand rule, and E&M problem-solving.
- Designed interactive worksheets and conceptual checkpoints to identify gaps in understanding and target high-impact physics topics.
- Fostered a collaborative environment where students learned to rely on peers, strengthening confidence and retention of course material.

Projects

Hellacious | *React.js, Flask, MongoDB, Firebase, AWS S3*

Oct 2025 – Present

- Developed a full-stack social media web app allowing users to post, like, and comment, with real-time updates and responsive design using React.js and Flask.
- Integrated Firebase Authentication for secure Google sign-in/log-in and connected Flask REST APIs to a MongoDB backend for user info.
- Configured AWS S3 for scalable image storage and optimized API routes for faster media retrieval.
- Used Git for version control and collaboration, and implemented unit tests to ensure backend reliability.

EcoRecyclr | *PyTorch, Kotlin*

Oct 2025 – Nov 2025

- Led a 4-member team during the Booz Allen Hamilton Code for Good hackathon and won 1st place for developing an Android app that classifies waste items to determine recyclability.
- Built an on-device ML model using PyTorch and fine-tuned ResNet-18 on 16,000+ labeled images, achieving 95% classification accuracy.
- Integrated the trained model into an Android app using Kotlin, implementing camera-based image capture and real-time predictions within the main Activity.
- Utilized background services to handle model inference and image preprocessing efficiently without blocking the UI thread, ensuring smooth user interaction.

Leadership/Extracurricular

Building STEPS

Jun 2024 – Jul 2024

- Tutored 15 high school juniors and seniors from Baltimore City high schools who aspire to go to college in math and algebra through the Building STEPS program, helping them improve their skills for college readiness.
- Maintained daily attendance logs to ensure consistent tracking and engagement

CWIT Retreat Planning Committee

Apr 2024 – Jul 2024

- Organized and facilitated icebreaker activities to foster connections among new scholars, ensuring a welcoming and collaborative environment.
- Developed and delivered presentation slides on the role of male scholars in CWIT, highlighting academic and professional expectations of the scholarship program.
- Supported an information session day by presenting personal research lab activities and assisting new scholars in showcasing their work.

Programming skills

Languages: C, C++, Python, Java, Kotlin, SQL, JavaScript, HTML/CSS, R

Frameworks: Flask, FastAPI, React.js, Node.js

Tools: Git, GitHub, VS Code, PyCharm